Install node js

What is Vite?

Vite is a **development tool/build tool** used to create and run modern web applications, including React applications.

In simple words:

Vite helps you create, start, and develop your React project quickly.

Why do we use Vite?

When you create a React project:

```
npm create vite@latest my-react-app
```

Vite creates the project structure for you.

Then when you run:

```
npm run dev
```

Vite starts a **development server** so you can open your React application in the browser.

Think of it like this

```
You write React code
        ↓
       Vite
        ↓
Development server
        ↓
    Browser
        ↓
React application
```

What does Vite do?

Vite mainly helps with:

1. **Creating the project**

```
npm create vite@latest my-react-app
```

2. **Running the development server**

```
npm run dev
```

3. **Processing your JavaScript/JSX and other files** while developing.

4. **Building the application for production**

```
npm run build
```

Vite is NOT React

This is important:

```
React → UI library
Vite  → Development/build tool
```

React is responsible for creating components:

```
function App() {
    return <h1>Hello World</h1>;
}
```

Vite helps you **develop and run** that React application.

Simple example

When you run:

```
npm create vite@latest my-react-app
```

you're basically saying:

"Vite, please create a new project for me."
When you run:

```
npm run dev
```

you're saying:
"Vite, start my project so I can see it in the browser."
In one line:
Vite is a fast tool that sets up, runs, and builds your React application during development.

1. **What does "Variant" mean?**

Variant means a particular version or option of something.
When Vite asks:

```
Select a variant:
› JavaScript
  JavaScript + SWC
  TypeScript
  TypeScript + SWC
```

it is asking:
"Which type of code/setup do you want to use for this React project?"
For a beginner
Choose:

```
JavaScript
```

This gives you a simple React project using JavaScript.
For example:

```
function App() {
    return <h1>Hello World</h1>;
}
```

Why are there different variants?
They provide different development setups:

Variant	Meaning
JavaScript	Normal JavaScript + React
JavaScript + SWC	JavaScript + React with SWC for faster compilation
TypeScript	TypeScript + React
TypeScript + SWC	TypeScript + React + SWC

Simple meaning

```
Framework → React
Variant   → How you want to use React
```

So when you select:

```
Framework: React
Variant: JavaScript
```

you are saying:
"Create a React project using JavaScript."

What is ESLint?
ESLint is a tool that **checks your JavaScript code for mistakes and bad coding practices**.
In simple words:
ESLint acts like a code checker for JavaScript and React.
Example

Suppose you write:

```
function App() {
    const name = "John";

    return <h1>Hello World</h1>;
}
```

You created `name`, but you never used it.
ESLint can detect this and warn you:

```
'name' is assigned a value but never used
```

Another example

You might accidentally write:

```
function App() {
    return <h1>Hello World</h1>
```

You forgot the closing `}`.
ESLint can identify syntax problems and report them.

Why is ESLint useful?

ESLint helps you:

- Find coding mistakes
- Find unused variables
- Maintain consistent coding style
- Identify potential problems
- Write cleaner JavaScript/React code

Where does ESLint come from?

When creating a Vite React project, you may see:

```
my-react-app
│
├── src
├── public
├── package.json
├── eslint.config.js
└── vite.config.js
```

The file:

```
eslint.config.js
```

contains ESLint configuration.

Important distinction

ESLint **does not run your React application**.

```
Vite
  ↓
Runs/develops your application

ESLint
  ↓
Checks your code
```

So:

```
npm run dev
```

➡ starts your React application.
And an ESLint command such as:

`npm run lint`

➡ checks your code for linting problems.

Simple definition

ESLint is a tool that analyzes JavaScript/React code and warns you about errors, potential bugs, and coding problems.

1. Create a React project Run: npm create vite@latest my-react-app You will get questions similar to: Project name: my-react-app Select a framework: React Select a variant: JavaScript Choose: React JavaScript Step 4: Go into the project Run: cd my-react-app Your terminal should now be inside the project folder. Step 5: Install dependencies Run: npm install explain these why they require
2. **ChatGPT said:**

Sure. These commands are required because they perform **different jobs** when creating and running a React project.

1. `npm create vite@latest my-react-app`

```
npm create vite@latest my-react-app
```

Why do we need this?

This command **creates a new React project** using Vite.

Break it down:

```
npm       → Node Package Manager
create    → Create a new project
vite      → Use Vite as the project tool
@latest   → Use the latest version
my-react-app → Name of your project
```

So this command basically means:

"Create a new project called `my-react-app` **using the latest Vite."**

It creates the basic files and folders for you.

For example:

```
my-react-app
│
├── public
├── src
├── index.html
├── package.json
└── vite.config.js
```

Without this command, you would have to manually create and configure a React project.

---

2. Select framework: React

Vite supports different frameworks.

For example:

```
React
Vue
Svelte
etc.
```

When you choose:

```
React
```

you are telling Vite:

"I want to build this project using React."

Vite then creates the appropriate React project structure and configuration.

---

3. Select variant: JavaScript

You may see:

```
JavaScript
JavaScript + SWC
TypeScript
TypeScript + SWC
```

If you are learning basic React, choose:

```
JavaScript
```

This means your React code will primarily use:

```
.js
.jsx
```

instead of TypeScript:

```
.ts
.tsx
```

For example:

```
function App() {
    return <h1>Hello World</h1>;
}
```

---

4. `cd my-react-app`

After creating the project, you run:

```
cd my-react-app
```

Why do we need this?

`cd` means:

Change Directory

Your terminal initially might be somewhere like:

```
C:\Users\YourName>
```

After:

```
cd my-react-app
```

you move into:

```
C:\Users\YourName\my-react-app>
```

This is important because the React project's files are inside that folder.

Think of it like entering a room before working with the things inside it.

```
Computer
  ↓
my-react-app folder
  ↓
package.json
src
index.html
```

...

---

5. `npm install`

Now you run:

`npm install`

Why do we need this?

The project contains a file called:

`package.json`

This file tells npm which packages the project needs.

For example, it contains dependencies similar to:

```
{
  "dependencies": {
    "react": "...",
    "react-dom": "..."
  },
  "devDependencies": {
    "vite": "..."
  }
}
```

When you run:

`npm install`

npm reads `package.json` and **downloads the required packages**.

They are placed inside:

`node_modules`

So:

```
package.json
      ↓
   npm install
      ↓
downloads required packages
      ↓
node_modules
```

Why can't we skip `npm install`?

Because your project needs React, React DOM, Vite, and other packages to work.

For example, your code may contain:

```
import { createRoot } from "react-dom/client";
```

The `react-dom` package must exist for that import to work.

Similarly, when you run:

`npm run dev`

you need Vite installed.

So `npm install` prepares all the required packages.

Complete process

Think of the commands like this:

```
1. npm create vite@latest my-react-app
                ↓
       Creates project
                ↓
2. Choose React
                ↓
       Uses React framework
                ↓
3. Choose JavaScript
                ↓
       Uses JavaScript
                ↓
4. cd my-react-app
                ↓
       Enter project folder
                ↓
5. npm install
                ↓
       Download dependencies
                ↓
6. npm run dev
                ↓
       Start development server
                ↓
7. Browser
                ↓
       React application
```

In very simple words

Command	Why?
`npm create vite@latest my-react-app`	Creates the React project
Choose **React**	Tells Vite to create a React application
Choose **JavaScript**	Use JavaScript instead of TypeScript
`cd my-react-app`	Enter the project folder
`npm install`	Download/install the packages the project needs
`npm run dev`	Start the React development server

The important distinction is:

`npm create vite` **creates the project structure.**
`npm install` **installs the project's required packages.**
`npm run dev` **actually starts the application.**

1. **React DOM in JavaScript**, it's the part of React that lets you render React components into the browser's actual DOM.

   Basic example

   ```
   import React from "react";
   import { createRoot } from "react-dom/client";

   function App() {
     return <h1>Hello, React!</h1>;
   }

   const root = createRoot(document.getElementById("root"));
   root.render(<App />);
   ```

   And your HTML might contain:

   ```
   <div id="root"></div>
   ```

   What happens?

1. `document.getElementById("root")` finds the HTML element.
2. `createRoot()` creates a React root.
3. `root.render(<App />)` tells React to display the `App` component inside that element.
4. React manages updates to the DOM efficiently.

   React DOM vs normal JavaScript

   Normal JavaScript:

   ```
   document.getElementById("root").innerHTML = "<h1>Hello</h1>";
   ```

   React:

   ```
   const root = createRoot(document.getElementById("root"));
   root.render(<h1>Hello</h1>);
   ```

   In modern React, you generally use `react-dom/client` and `createRoot()` rather than the older `ReactDOM.render()` API.

2. **You said:**

   give some and execuetion process

3. **ChatGPT said:**

   Sure. Let's understand **React DOM with an example and its execution process step by step**.

   Example

   ```
   import { createRoot } from "react-dom/client";

   function App() {
     return <h1>Hello World</h1>;
   }

   const rootElement = document.getElementById("root");

   const root = createRoot(rootElement);

   root.render(<App />);
   ```

   HTML:

```
<!DOCTYPE html>
<html>
  <body>
    <div id="root"></div>

    <script type="module" src="index.js"></script>
  </body>
</html>
```

Execution process

Think of the process like this:

Browser → HTML → JavaScript → React DOM → Component → DOM

1. Browser loads HTML

The browser sees:

```
<div id="root"></div>
```

At this point, the page has an empty `div`.

---

2. JavaScript file executes

The browser loads `index.js`.

This code runs:

```
const rootElement = document.getElementById("root");
```

JavaScript finds the HTML element:

```
<div id="root"></div>
```

So:

```
rootElement
    ↓
<div id="root"></div>
```

---

3. `createRoot()` creates a React root

```
const root = createRoot(rootElement);
```

React now knows:

"I should manage the UI inside this `<div>`."

Conceptually:

```
React
  ↓
root
  ↓
<div id="root"></div>
```

---

4. React component is created

We have:

```
function App() {
  return <h1>Hello World</h1>;
}
```

And:

```
root.render(<App />);
```

React executes the `App` component.

Conceptually:

```
App()
```

```
 ↓
<h1>Hello World</h1>
```

---

5. React updates the DOM

React takes the result:

```
<h1>Hello World</h1>
```

and places it inside:

```
<div id="root">
  <h1>Hello World</h1>
</div>
```

The browser then displays:

Hello World

More interesting example

```
import { createRoot } from "react-dom/client";

function App() {
  const name = "John";

  return (
    <div>
      <h1>Hello {name}</h1>
      <p>Welcome to React</p>
    </div>
  );
}

const root = createRoot(document.getElementById("root"));

root.render(<App />);
```

Execution:

```
index.js starts
      ↓
find #root
      ↓
createRoot(#root)
      ↓
root.render(<App />)
      ↓
App() executes
      ↓
JSX is converted into React elements
      ↓
React creates/updates DOM
      ↓
Browser displays:
      Hello John
      Welcome to React
```

Important point

`createRoot()` **does not directly create the HTML you see**. It creates a React root that allows React to manage the DOM.

The key line is:

```
root.render(<App />);
```

This tells React:

"Render the `App` component into this root."

If you want, I can next explain **Virtual DOM → Reconciliation → Real DOM** with a simple example, which is the most important part for understanding how React actually works.

4. **You said:**

theory

5. **ChatGPT said:**

React DOM — Theory

React DOM is a package that provides methods to connect **React** with the browser's **DOM (Document Object Model)**.

In simple words:

React creates and manages the user interface, while React DOM connects that React interface to the actual HTML DOM in the browser.

Main concept

When a React application starts, the browser first loads an HTML page containing a root element:

```
<div id="root"></div>
```

React then uses **React DOM** to take control of this root element.

The modern way is:

```
import { createRoot } from "react-dom/client";

const root = createRoot(document.getElementById("root"));

root.render(<App />);
```

Execution process

The basic execution flow is:

1. Browser loads HTML

The browser creates the DOM tree.

2. JavaScript loads

The React application's JavaScript code starts executing.

3. React finds the root element

React gets the element using:

```
document.getElementById("root")
```

4. `createRoot()` creates a React root

React establishes a connection between the React application and that DOM element.

5. `render()` is called

React receives the component that should be displayed.

```
root.render(<App />);
```

6. Component executes

React calls the component and gets the UI it should represent.

7. React creates a virtual representation
React internally represents the desired UI and compares it with the previous UI when updates occur.
8. Reconciliation happens
When the application changes, React determines what parts of the UI actually need to change.
9. Real DOM is updated
React DOM applies the necessary changes to the browser's actual DOM.
10. Browser displays the result
The user sees the updated interface.
Simple flow

```
HTML
  ↓
Browser DOM
  ↓
React JavaScript
  ↓
createRoot()
  ↓
root.render()
  ↓
Component
  ↓
React's UI representation
  ↓
Reconciliation
  ↓
React DOM
  ↓
Real DOM
  ↓
Browser UI
```

Why React DOM is useful
React DOM helps React:

- Render React components into HTML.
- Update the browser DOM efficiently.
- Handle UI changes when state or props change.
- Connect React's component system with the browser environment.

React vs React DOM
React mainly provides the core functionality for creating components and managing the UI.
React DOM provides the functionality needed to render that UI into the **browser DOM**.
So, conceptually:

```
React     → Creates/describes UI
React DOM → Puts that UI into the browser DOM
```

That's why you commonly see:

```
import { createRoot } from "react-dom/client";
```
in a React web application's entry point.

React DOM

React DOM is a part of React that helps **React display and update components in the browser's HTML DOM**.
In simple words:
React creates the UI, and React DOM puts that UI into the web page.
What is DOM?
DOM stands for **Document Object Model**.
When the browser loads HTML like:

the browser converts it into a DOM structure that JavaScript can access and modify.
What does React DOM do?
React DOM connects your React application to that DOM.
For example:

```
import { createRoot } from "react-dom/client";

const root = createRoot(document.getElementById("root"));

root.render(<App />);
```

Here:

- document.getElementById("root") → finds the HTML element.
- createRoot() → creates a React root.
- root.render() → displays the React component inside that root.

Simple execution flow
HTML
↓

↓
React DOM finds "root"
↓
createRoot()
↓
render(<App />)
↓
React component
↓
Browser DOM gets updated
↓
User sees the UI
Why is React DOM important?

Without React DOM, a React application wouldn't know **where to display its components in a browser's DOM**.
For example:
React
↓
Creates UI
↓
React DOM
↓
Connects UI to browser DOM
↓
Browser
↓
Displays UI

React vs React DOM

React	React DOM
Creates components and UI	Connects React UI to browser DOM
Handles component logic	Renders UI into the DOM
Used to describe what UI should look like	Used to display it in a web page

Definition
React DOM is the package that allows React applications to render and update React components in the browser's DOM.

What does render **mean in React?**
Render means "display something on the screen."
In React, when we write:
root.render(<App />);
it means:
"Display the App **component inside the root element."**

Simple example
Suppose App component is:

```
function App() {
   return <h1>Hello World</h1>;
}
```

Then:
root.render(<App />);
tells React to display:
<h1>Hello World</h1>
inside:

So the browser ends up with:

```
  <h1>Hello World</h1>
</div>
```
And you see:
Hello World
In simple words
render = display
So:
```
root.render(<App />);
```
means:
"Display the App component on the webpage."

What is an App **Component in React?**
The **App component** is usually the **main/root component** of a React application.
A component is basically a **JavaScript function that returns UI**.
For example:
```
function App() {
  return <h1>Hello World</h1>;
}
```
Here:

App → component name

function App() → defines the component

return → tells React what UI to display

<h1>Hello World</h1> → the UI

How it works
Usually, App is rendered from main.jsx:
```
import { createRoot } from "react-dom/client";
import App from "./App.jsx";

const root = createRoot(document.getElementById("root"));

root.render(<App />);
```
The execution is:
main.jsx
  ↓
<App />
  ↓
App() executes
  ↓
return <h1>Hello World</h1>
  ↓
React DOM

```
  ↓
<div id="root">
   <h1>Hello World</h1>
</div>
  ↓
```
Browser displays "Hello World"

Why is it called App**?**

App is **just a common name**. It is not a special JavaScript keyword.

We can also write:

```
function MyApplication() {
   return <h1>Hello World</h1>;
}
```

and then:

```
root.render(<MyApplication />);
```

It works the same way.

A component is a reusable piece of UI, and App **is commonly used as the main component of a React application.**

Program

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>React App</title>
</head>

<body>
    <div id="root"></div>

    <script type="module" src="/src/main.jsx"></script>
</body>
</html>
```

src/main.jsx

```
import { createRoot } from "react-dom/client";
import App from "./App.jsx";

const root = createRoot(document.getElementById("root"));

root.render(<App />);
```

src/App.jsx

```
function App() {
    return (
        <div>
            <h1>Hello World</h1>
            <p>My First React Application</p>
        </div>
    );
}

export default App;
```

Step-by-step execution

Step 1: create the React project

In the VS Code terminal:

```
npm create vite@latest my-react-app  // note if npm doesn't work type npm.cmd
create vite@latest my-react-app
```

select:

```
React
JavaScript
```

Vite creates the project structure.

```
my-react-app
│
├── index.html
├── package.json
├── node_modules
└── src
    ├── App.jsx
    └── main.jsx
```

Step 2: Install packages

You run:

```
npm install
```

`npm` reads `package.json` and downloads the required packages into:

```
node_modules
```

Among these packages are React and React DOM.

Conceptually:

```
package.json
      ↓
npm install
      ↓
node_modules
      ↓
React + React DOM + Vite + other packages
```

Step 3: Start the application

You run:

```
npm run dev
```

What does this do?

`npm` looks inside `package.json`.

You will have a script similar to:

```
"scripts": {
    "dev": "vite"
}
```

Therefore:

```
npm run dev
     ↓
npm reads package.json
     ↓
finds "dev": "vite"
     ↓
starts Vite
```

Vite starts a development server.

You get something like:

```
Local: http://localhost:5173/
```

Step 4: Open the URL

You open:

```
http://localhost:5173/
```

The browser sends a request to the Vite development server.

```
Browser
   ↓
http://localhost:5173/
   ↓
Vite
```

Vite sends your `index.html` to the browser.

Step 5: Browser reads `index.html`

The browser starts executing the HTML.

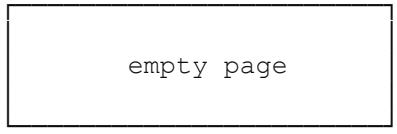
It finds:

```
<div id="root"></div>
```

The browser creates this DOM:

```
HTML
 └── BODY
      └── DIV#root
```

At this moment, the page is basically empty.



Because:

```
<div id="root"></div>
```

doesn't contain anything yet.

Step 6: Browser finds the script

The browser then sees:

```
<script type="module" src="/src/main.jsx"></script>
```

This tells the browser:

Load and execute `main.jsx`.

So the browser requests:

```
/src/main.jsx
```

from Vite.

```
Browser
   ↓
main.jsx
   ↓
Vite
   ↓
JavaScript returned to browser
```

Step 7: main.jsx **starts executing**

The first line is:

```
import { createRoot } from "react-dom/client";
```

This imports `createRoot()` from React DOM.

Conceptually:

```
React DOM
   ↓
createRoot()
   ↓
available inside main.jsx
```

Step 8: Import App

Next:

```
import App from "./App.jsx";
```

This loads your `App.jsx` file.

The browser/Vite finds:

```
src/App.jsx
```

Inside it:

```
function App() {
    return (
        <div>
            <h1>Hello World</h1>
            <p>My First React Application</p>
        </div>
    );
}

export default App;
```

The `App` function is loaded.

At this point, **the component is defined**.

It hasn't been rendered yet.

Step 9: Find the root **element**

Now this line executes:

```
document.getElementById("root")
```

Remember that the browser already created:

```
<div id="root"></div>
```

JavaScript searches the DOM for the element whose ID is `"root"`.
It finds it.
Conceptually:

```
document
   ↓
getElementById("root")
   ↓
<div id="root"></div>
```

The result is stored here:

```
const rootElement = document.getElementById("root");
```

In our shorter code, it's passed directly to `createRoot()`.

Step 10: createRoot() **executes**

This line now runs:

```
const root = createRoot(document.getElementById("root"));
```

React creates a **React root** connected to:

```
<div id="root"></div>
```

Think of it as:

```
React
  │
  ↓
React Root
  │
  ↓
<div id="root"></div>
```

React knows:
It is responsible for managing the UI inside this element."

Step 11: root.render() **executes**

Now we reach the most important line:

```
root.render(<App />);
```

This means:
Render the `App` component inside the React root.
So React receives:

```
<App />
```

Step 12: React executes App

React needs to know what `<App />` should produce.
So the `App` component function executes:

```
function App() {
    return (
        <div>
            <h1>Hello World</h1>
            <p>My First React Application</p>
        </div>
    );
}
```

The result is conceptually:

```
<div>
    <h1>Hello World</h1>
    <p>My First React Application</p>
</div>
```
So:
```
<App />
   ↓
App()
   ↓
<div>
   <h1>Hello World</h1>
   <p>My First React Application</p>
</div>
```

Step 13: JSX is processed

```
<h1>Hello World</h1>
```
This is **JSX**.
JSX allows us to write HTML-like syntax inside JavaScript.
The build process transforms JSX into JavaScript representing React elements.
Conceptually:
```
JSX
 ↓
JavaScript
 ↓
React element representation
```

Step 14: React determines the required UI

React now knows that the UI should contain:
```
div
 ├── h1 → Hello World
 └── p  → My First React Application
```
React prepares the corresponding DOM changes.

Step 15: React DOM updates the browser DOM

Initially the browser had:
```
<div id="root"></div>
```
React DOM updates it to effectively contain:
```
<div id="root">
    <div>
        <h1>Hello World</h1>
        <p>My First React Application</p>
    </div>
</div>
```
This is the connection between **React** and the browser's **real DOM**.

Step 16: Browser displays the result

The browser now paints the updated DOM on the screen.
You see:

```
Hello World

My First React Application
```

React application is running.

Complete execution flow

```
1. npm create vite
        ↓
2. npm install
        ↓
3. npm run dev
        ↓
4. Vite starts development server
        ↓
5. Browser opens localhost:5173
        ↓
6. Browser loads index.html
        ↓
7. Browser creates <div id="root">
        ↓
8. Browser loads main.jsx
        ↓
9. main.jsx imports React DOM
        ↓
10. main.jsx imports App.jsx
        ↓
11. document.getElementById("root")
        ↓
12. createRoot()
        ↓
13. root.render(<App />)
        ↓
14. App() executes
        ↓
15. App returns JSX
        ↓
16. JSX is processed
        ↓
17. React determines UI changes
        ↓
18. React DOM updates Real DOM
        ↓
19. Browser displays the UI
```

1. Find HTML element

```
document.getElementById("root")
```

Meaning: Find the HTML element where React should work.

2. Create React root

```
createRoot(document.getElementById("root"))
```

Meaning: Tell React to manage that DOM element.

3. Render component

```
root.render(<App />);
```
Meaning: Tell React which component/UI to display.
So the core idea is:
```
HTML root
   ↓
createRoot()
   ↓
render()
   ↓
React Component
   ↓
Real DOM
   ↓
Browser Screen
```
This is the basic **React DOM execution process**.